

Guidelines: Fever in Infants <90 days old

INTRODUCTION

- o The main aim when faced with a febrile infant < 90 days old is to accurately exclude Serious Bacterial Infections (SBIs) through risk stratification.
- o Clinical judgement alone is not reliable to pick up these infants with SBIs as they often show no localising signs of clinical illness, especially in the neonatal group.
- o The big SBIs not to miss are meningitis, urinary tract infections, bacteraemia, pneumonia, bone and soft tissue infections and bacterial gastroenteritis.
- o More recent emphasis on risk stratifying for Invasive Bacteria Infections (IBIs) comprising of meningitis and bacteraemia. Clinical suspicion of IBIs in febrile infants warrant admission for treatment with intravenous antibiotics and close monitoring.

MODIFIED ROCHESTER CRITERIA

To risk stratify patients into HIGH and LOW risk for Invasive Bacterial Infections:

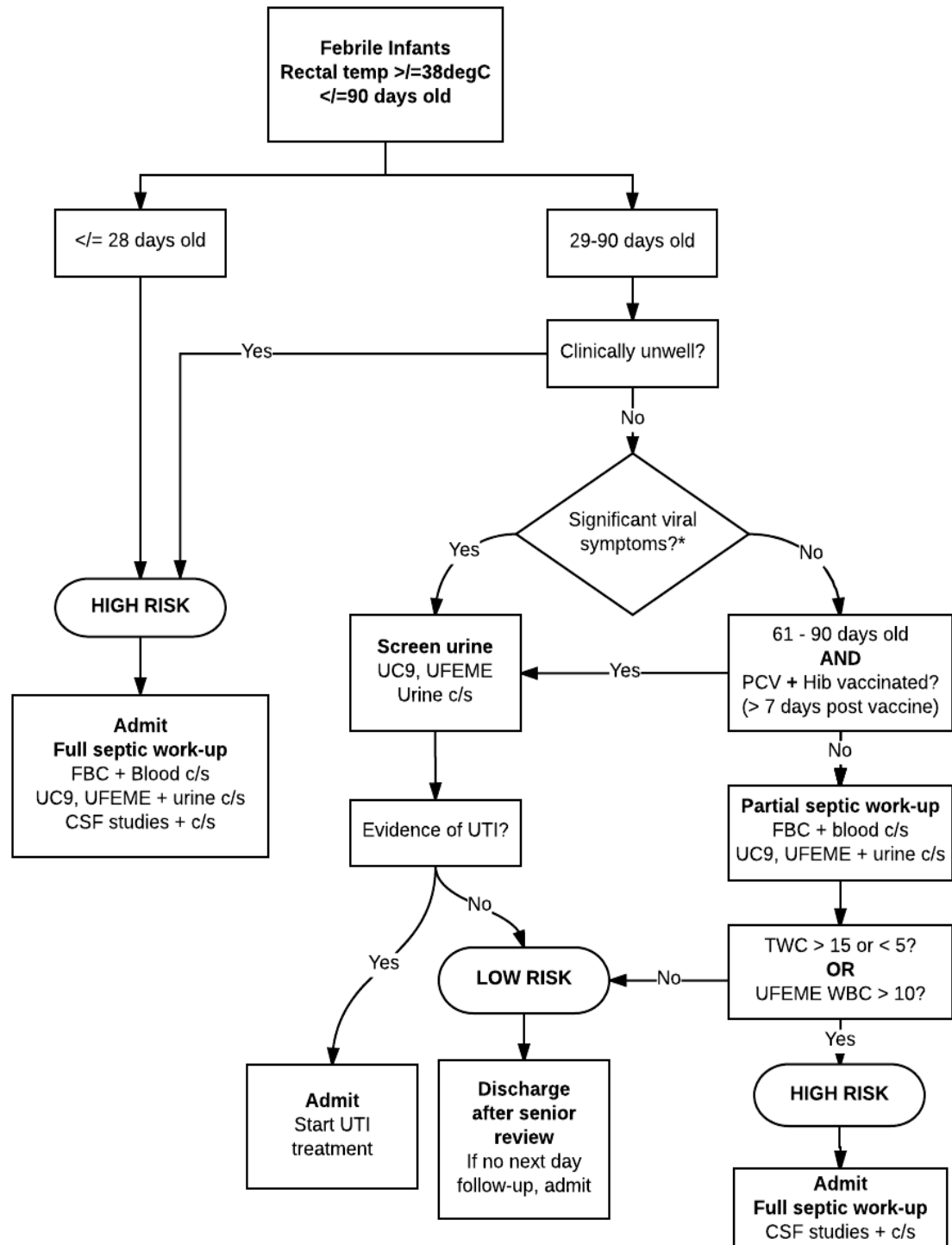
- o Neonates (0 – 28 day old) with fever are at high risk of invasive bacterial infections requiring a full septic work-up including blood, urine and CSF studies and initiation of empiric antibiotics/antivirals.
- o In the post-vaccination era for *Pneumococcus* and *Haemophilus influenzae*, the risk of meningitis is very low in the post-neonatal group.

*Febrile young Infants with significant viral infections**

- o Significant viral symptoms in febrile young infants aged 29-90 days old reduce their risk of having non-UTI SBIs.
- o However, the rate of concomitant urinary tract infection is still significant in these post-neonatal infants with viral symptoms
 - In-out urinary catheterisation for UC9, urine formed elements and urine culture should be done in these young infants.

* Infants < 3 months old with clinical evidence of bronchiolitis should be managed as per Department Guidelines for Bronchiolitis. Admission for monitoring of respiratory status, apnoeas and feeding is warranted.

CLINICAL FLOW DIAGRAM OF MODIFIED ROCHESTER CRITERIA



*Significant viral symptoms include respiratory symptoms like prominent nasal congestion, wheeze, crepitations, tachypnoea and respiratory distress.

FOR ALL HIGH RISK PATIENTS

Full septic work up includes:

- o Blood tests
 - o Full Blood Count
 - o Blood aerobic culture
- o Urine tests
 - o UC9^a
 - o Urine Formed Elements
 - o Urine culture^b
- o Cerebrospinal fluid tests^c
 - o CSF panel
 - o Cell count/cytospin
 - o CSF culture
 - o ± Viral studies (HSV, Enterovirus) as clinically indicated
- o CXR (with presence of pulmonary symptoms)
- o Stool microscopy and cultures (with symptoms of gastroenteritis)

a. High specificity for urinary tract infection (Rules in a UTI if nitrites +, leucocytes +)

b. Only in-out catheterisation samples accurately reflect presence of urinary tract infection on urine culture (>10,000 organisms = Urinary tract infection)

c. Lumbar punctures should be performed for all High Risk patients

- o Neonates <28 days old
- o Ill-appearing
- o Seizures
- o Laboratory criteria as per Modified Rochester Criteria for High Risk stratification
- o Prior to administration of empiric antibiotics

Inflammatory markers (CRP, procalcitonin) are not routinely performed as part of the full septic work up. They have high specificities but low sensitivities, and are therefore not a very useful screening tool.

TREATMENT OF HIGH RISK FEBRILE INFANTS

1. Treat with empiric antibiotics IV Ampicillin and IV Gentamicin (See below for dosing recommendations) after full septic work-up.
2. Add IV Acyclovir for infants with:
 - o Ill-appearance
 - o Muco-cutaneous vesicles, seizures (manifestations of HSV infection)
 - o Maternal history of active HSV infection
 - o Transaminitis in liver panel
 - o CSF cell-count with high RBC in an atraumatic tap
3. Continue empiric treatment until culture and sensitivities are out to rationalise antibiotic choice.

ANTIMICROBIAL DOSING GUIDE

Ampicillin - 50mg/kg/dose by IV slow push, or IM

PMA (weeks)	Post-natal (days)	Interval (hours)
≤ 29	0 to 28	12
	> 28	8
30 to 36	0 to 14	12
	> 14	8
37 to 44	0 to 7	12
	> 7	8
≥ 45	ALL	6

Gentamicin

PMA (weeks)	Post-natal (days)	Dose (mg/kg)	Interval (hours)
≤ 29*	0 to 7	5	48
	8 to 28	4	36
	≥ 29	4	24
30 to 34	0 to 7	4.5	36
	≥ 8	4	24
35 to 37	ALL	4	24
≥ 37	0 to 28	4	24
	≥ 29	6	24

**or significant asphyxia, PDA, or treatment with indomethacin/brufen*

Acyclovir

20mg/kg/dose Q8H

IV infusion by syringe pump over 1 hour

Prolong the dosing interval in premature infants <34 weeks PMA or in patients with significant renal impairment or hepatic failure.

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